

ANNEX III

HYDRAULIC DESIGN

Table 1: Determination of Runoff coefficient

Region	Area (ha)	coefficient of runoff	Remarks	A*C
1	1.1	0.15	Mix of houses and soil	0.165
2	0.5	0.25	vegetation and houses	0.125
3	0.11	0.25	vegetation and houses	0.0275
4	0.1	0.3	houses	0.03
5	0.13	0.3	houses	0.039
6	0.31	0.1	slope	0.031
7	0.35	0.1	slope with vegetation	0.035
8	0.1	0.1	slope	0.01
9	0.5	0.25	houses and vegetation	0.125
10	0.35	0.3	houses	0.105
11	0.62	0.3	houses	0.186
12	0.35	0.25	houses and vegetation	0.0875
13	0.4	0.1	slope	0.04
14	0.0398	0.8	concrete	0.03184
15	0.11	0.3	houses	0.033
16	0.23	0.1	slope	0.023
17	0.14	0.1	slope and vegetation	0.014
18	1.32	0.25	houses and vegetation	0.33
19	0.1	0.15	slope and vegetation	0.015
20	0.58	0.15	slope and vegetation	0.087
21	0.21	0.1	slope	0.021
22	0.15	0.1	slope	0.015
23	0.29	0.25	houses and vegetation	0.0725
24	0.25	0.15	slope and vegetation	0.0375
25	0.42	0.25	vegetation and houses	0.105
26	0.17	0.3	houses	0.051
27	0.1	0.3	houses	0.03
28	6.1	0.15	slope and vegetation	0.915
29	7.55	0.15	slope and vegetation	1.1325
30	22.1	0.15	slope and vegetation	3.315
31	8.63	0.05	slope and vegetation at ridge	0.4315
32	12.2	0.05	slope and vegetation at ridge	0.61
Sum	65.61			8.276

The coefficient of runoff was determined as 0.126137.

Table 2: Annual maxima series (AMS)

Year (A.D.)	Max of Rainfall (mm)
1980	95.5
1981	79.3
1982	69
1983	72.5
1984	85
1985	75
1986	179.4
1987	90.6
1988	72.4
1989	97.6
1990	101.2
1991	92.5
1992	86.8
1993	75
1994	98
1995	95.8
1996	114.9
1997	124.9
1998	64.7
1999	89
2000	64
2001	82
2002	161.5
2003	115.7
2004	96.2
2005	113.5
2006	67.4
2007	78
2008	84
2009	104.5
2010	109
2011	83.4
2012	125.6
2013	71.6
2014	61
2015	84
2016	70
2017	74
2018	117
2019	83.6
2020	61
2021	116.4
2022	65.2
Grand Total	179.4
Average (mm)	91.80697674
Standard Deviation(mm)	25.24941673

Table 3: Standards of Y_n & S_n for Gumbel's Method

Reduced mean and standard deviation		
No, of years of record (n)	Reduced mean, Y_n	Reduced standard deviation, S_n
8	0.4843	0.9043
9	0.4902	0.9188
10	0.4952	0.9497
11	0.4996	0.9676
12	0.5035	0.9833
13	0.5070	0.9972
14	0.5100	1.0095
15	0.5128	1.0206
16	0.5157	1.0316
17	0.5181	1.0411
18	0.5202	1.0493
19	0.5220	1.0566
20	0.5236	1.0628
21	0.5252	1.0696
22	0.5268	1.0754
23	0.5283	1.0811
24	0.5296	1.0864
25	0.5309	1.0915
26	0.5320	1.0961
27	0.5332	1.1004
28	0.5343	1.1047
29	0.5353	1.1086
30	0.5362	1.1124
31	0.5371	1.1159
32	0.5380	1.1193
33	0.5388	1.1226
34	0.5396	1.1255
35	0.5403	1.1285
36	0.5410	1.1313
37	0.5418	1.1339
38	0.5424	1.1363
39	0.5430	1.1388
40	0.5436	1.1413
41	0.5442	1.1436
42	0.5448	1.1458
→ 43	0.5453	1.1480
44	0.5458	1.1499
45	0.5463	1.1519
46	0.5468	1.1538
47	0.5473	1.1557
48	0.5477	1.1574
49	0.5481	1.1590

Reduced mean and standard deviation		
No, of years of record (n)	Reduced mean, Y_n	Reduced standard deviation, S_n
50	0.5485	1.1607
51	0.5489	1.1623
52	0.5493	1.1638
53	0.5497	1.1638
54	0.5501	1.1667
55	0.5504	1.1681
56	0.5508	1.1696
57	0.5511	1.1708
58	0.5515	1.1721
59	0.5518	1.1734
60	0.5521	1.1747

DETERMINATION OF INTENSITY OF RAINFALL:

<u>Analysis for max 24 hours rainfall using Gumbel 's Type I Extreme Distribution Function</u>			
Average	<i>Qavg</i>	91.81	mm
Standard Deviation	<i>s</i>	25.249	mm
Record Length	<i>n</i>	43	
Mean of Gumbel Variate	<i>Yn</i>	0.5453	mm
Std Dev of Gumbel Variate	<i>Sn</i>	1.148	mm

Return period (T)	Reduced Variate (Y_t)	Frequency Factor (K)	Maximum 24 Hr. Rainfall (R_{24}), mm	Remarks
2.000	0.367	-0.156	87.868	Side drain
5.000	1.500	0.832	112.804	
10.000	2.250	1.485	129.309	
20.000	2.970	2.112	145.141	
25.000	3.199	2.311	150.163	Cross structure
50.000	3.902	2.924	165.634	
100.000	4.600	3.532	180.990	

The return period for side drain is taken as 2 yrs and for cross drainage structures is taken as 25 yrs.

Rainfall intensity for side drain=	76.766	mm/hr
Rainfall intensity for cross structure=	131.18	mm/hr

FORMULAE USED IN ABOVE CALCULATION:

Reduced variate(Y) = $-\ln \ln (-\ln \ln (T-1/T))$

- Frequency factor (K)= $Y - Y_n S_n$
- Maximum 24 hr. rainfall (R24)= $\bar{x} + k * \sigma$

Intensity (I)= $(R24/24) * (24/t)^{2/3}$

Let rainfall duration be 't' = 15 min = 15/60 hr. = 0.25 hr.

Initially the default cross section dimensions of the drain in SW Road software are:

For Type-G drain:			For Type-F drain		
Bottom Width (B)=	0.475	m	A	0.15	m ²
Hill side slope= 1H:0V			P	1.344	m
Road side slope(Z)= 1H:0.5V	0.5		R	0.111607	m
Depth (D)=	0.3 m				
Top width (Tw)=	0.625	m			
Cover thickness (t _c)=	0.2	m			

SAMPLE CALCULATION OF FULL FLOW IN SIDE DRAIN G-TYPE:

$$\text{Area of drain}(A) = BD + \left(\frac{1}{2} * D * (0.625 - 0.475)\right) = 0.165 \text{ m}^2$$

$$\text{Perimeter}(P) = (B + D + Tw + D\sqrt{1 + Z^2}) = 1.73541 \text{ m}$$

$$R = \frac{A}{P} = 0.09508 \text{ m}$$

Then, from manning's equation:

Assuming manning's roughness coefficient (n)= 0.013,

Considering the longitudinal slope of road at a section (S) = 0.0355

$$\text{Velocity}(v) = \frac{1}{n} * R^{\frac{2}{3}} * S^{\frac{1}{2}} = 3.019201 \text{ m/s}$$

$$Q_{\text{drainage capacity}} = A * v = 0.4981682 \text{ m}^3/\text{s}$$

SIDE DRAIN TABLES

SLOPE (%)	Slope	V _{dG}	V _{dF}	Q _{d(Drain-F)}	Q _{d(Drain-G)}	Q _s	Check Q _d > Q _s	If Q _d < Q _s , Take Q _d = Q _s	Slope (DEG)	Vertical Elevation (m)	Horizontal Length(m)	Catchment Area(A), (m ²)	Offset	Remark	CHAINAGE (m)	Qd/QS	Limit Qd/Qs to (10-20%)	Cummulative flow left right		Culvert or not
3.55	0.0355	3.019	3.360	0.504	0.498	0.005	TRUE	0.498	2.033	3.103	87.399	1747.980	20	left	87.399	105.958	Change size	0.033	0.020	-
						0.006	TRUE	0.498				2184.975	25	right		84.766	Change size	0.033	0.020	-
1.82	0.0182	2.162	2.406	0.361	0.357	0.010	TRUE	0.357	1.043	2.995	164.535	0.000		Provided Same as min slope=1.38%	251.934	-		0.029	0.014	-
						0.005	TRUE	0.357				0.000				-		0.029	0.014	-
1.55	0.0155	1.995	2.220	0.333	0.329	0.010	TRUE	0.329	0.888	2.565	165.478	0.000			417.412	-		0.019	0.010	-
						0.005	TRUE	0.329				0.000				-		0.019	0.010	-
1.38	0.0138	1.882	2.095	0.314	0.311	0.010	TRUE	0.311	0.791	1.640	118.808	3564.240	30	left	536.220	32.399	Change size	0.010	0.005	-
						0.005	TRUE	0.311				1782.120	15	right		64.797	Change size	0.010	0.005	-
-3.02	-0.0302	2.785	3.099	0.465	0.459	0.019	TRUE	0.459	(1.730)	4.313	142.811	7140.550	50	left	679.031	23.924	Change size	0.256	0.043	slope change
						0.004	TRUE	0.459				1428.110	10	right		119.618	Change size	0.256	0.043	-
1.12	0.0112	1.696	1.887	0.283	0.280	0.020	TRUE	0.280	0.642	1.659	148.152	7407.600	50	left	827.183	14.044	ECONOMICAL	0.237	0.040	causeway
						0.012	TRUE	0.280				4444.560	30	right		23.406	Change size	0.237	0.040	-
4.05	0.0405	3.225	3.588	0.538	0.532	0.006	TRUE	0.532	2.319	3.153	77.849	2335.470	30	left	905.032	84.705	Change size	0.217	0.028	-
						0.004	TRUE	0.532				1556.980	20	right		127.057	Change size	0.217	0.028	-
9.99	0.0999	5.065	5.636	0.845	0.836	0.012	TRUE	0.836	5.705	14.963	149.783	4493.490	30	left	1054.815	69.144	Change size	0.211	0.023	-
						0.008	TRUE	0.836				2995.660	20	right		103.716	Change size	0.211	0.023	-
5.63	0.0563	3.802	4.231	0.635	0.627	0.004	TRUE	0.627	3.222	8.064	143.238	1432.380	10	left	1198.053	162.836	Change size	0.199	0.015	-
						0.015	TRUE	0.627				5729.520	40	right		40.709	Change size	0.199	0.015	-
9.96	0.0996	5.057	5.627	0.844	0.834	0.002	TRUE	0.834	5.688	7.984	80.162	801.620	10	left	1278.215	387.004	Change size	0.195	0.059	culvert
						0.004	TRUE	0.834				1603.240	20	right		193.502	Change size	0.195	0.059	-
3.22	0.0322	2.875	3.200	0.480	0.474	0.002	TRUE	0.474	1.844	2.834	88.006	880.060	10	left	1366.221	200.433	Change size	0.193	0.055	-
						0.005	TRUE	0.474				1760.120	20	right		100.217	Change size	0.193	0.055	-
8.41	0.0841	4.647	5.171	0.776	0.767	0.010	TRUE	0.767	4.807	10.073	119.774	3593.220	30	left	1485.995	79.336	Change size	0.191	0.050	-
						0.003	TRUE	0.767				1197.740	10	right		238.007	Change size	0.191	0.050	-
10	0.1	5.067	5.639	0.846	0.836	0.005	TRUE	0.836	5.711	12.069	120.689	1810.335	15	left	1606.684	171.710	Change size	0.181	0.047	-
						0.010	TRUE	0.836				3620.670	30	right		85.855	Change size	0.181	0.047	-
0.56	0.0056	1.199	1.334	0.200	0.198	0.005	TRUE	0.198	0.321	0.524	93.641	1872.820	20	left	1700.325	39.278	Change size	0.176	0.037	-
						0.005	TRUE	0.198				1872.820	20	right		39.278	Change size	0.176	0.037	-
6.97	0.0697	4.231	4.708	0.706	0.698	0.006	TRUE	0.698	3.987	8.390	120.376	2407.520	20	left	1820.701	107.796	Change size	0.171	0.032	-
						0.006	TRUE	0.698				2407.520	20	right		107.796	Change size	0.171	0.032	-
3.73	0.0373	3.095	3.444	0.517	0.511	0.005	TRUE	0.511	2.136	3.644	97.688	1953.760	20	left	1918.389	97.171	Change size	0.165	0.025	-
						0.003	TRUE	0.511				976.880	10	right		194.342	Change size	0.165	0.025	-
7.23	0.0723	4.309	4.795	0.719	0.711	0.006	TRUE	0.711	4.135	8.149	112.716	2254.320	20	left	2031.105	117.249	Change size	0.159	0.023	-
						0.003	TRUE	0.711				1127.160	10	right		234.497	Change size	0.159	0.023	-
2.31	0.0231	2.435	2.710	0.407	0.402	0.004	TRUE	0.402	1.323	1.592	68.931	1378.620	20	left	2100.036	108.372	Change size	0.153	0.020	-
						0.002	TRUE	0.402				689.310	10	right		216.743	Change size	0.153	0.020	-
4.54	0.0454	3.414	3.799	0.570	0.563	0.003	TRUE	0.563	2.599	4.680	103.092	1030.920	10	left	2203.128	203.169	Change size	0.149	0.018	-
						0.003	TRUE	0.563				1030.920	10	right		203.169	Change size	0.149	0.018	-
3.98	0.0398	3.197	3.557	0.534	0.527	0.002	TRUE	0.527	2.279	3.209	80.624	806.240	10	left	2283.752	243.238	Change size	0.147	0.015	-
						0.002	TRUE	0.527				806.240	10	right		243.238	Change size	0.147	0.015	-
11.9	0.119	5.528	6.151	0.923	0.912	0.039	TRUE	0.912	6.786	56.942	478.508	14355.240	30	left	2762.260	23.622	Change size	0.145	0.013	-
						0.013	TRUE	0.912				4785.080	10	right		70.866	Change size	0.145	0.013	-
9.47	0.0947	4.931	5.487	0.823	0.814	0.002	TRUE	0.814	5.410	8.597	90.780	907.800	10	left	2853.040	333.226	Change size	0.106	0.010	culvert
						0.005	TRUE	0.814				1815.600	20	right		166.613	Change size	0.106	0.010	-

SLOPE (%)	Slope	V _{dG}	V _{dF}	Q _{d(Drain-F)}	Q _{d(Drain-G)}	Q _s	Check Q _d > Q _s	If Q _d < Q _s , Take Q _d = Q _s	Slope (DEG)	Vertical Elevation (m)	Horizontal Length(m)	Catchment Area(A), (m ²)	Offset	Remark	CHAINAGE (m)	Qd/QS	Limit Qd/Qs to (10-20%)	Cummulative flow left right		Culvert or not
14.87	0.1487	6.179	6.876	1.031	1.020	0.015	TRUE	1.020	8.458	27.294	183.550	5506.500	30	left	3036.590	68.839	Change size	0.104	0.005	-
						0.005	TRUE	1.020				1835.500	10	right		206.517	Change size	0.104	0.005	-
10.79	0.1079	5.264	5.857	0.879	0.869	0.031	TRUE	0.869	6.158	49.123	455.263	11381.575	25	left	3491.853	28.370	Change size	0.089	0.053	culvert
						0.018	TRUE	0.869				6828.945	15	right		47.284	Change size	0.089	0.053	-
8.12	0.0812	4.566	5.081	0.762	0.753	0.011	TRUE	0.753	4.642	8.013	98.685	3947.400	40	left	3590.538	70.961	Change size	0.058	0.035	-
						0.003	TRUE	0.753				986.850	10	right		283.845	Change size	0.058	0.035	-
9.95	0.0995	5.055	5.625	0.844	0.834	0.028	TRUE	0.834	5.682	25.587	257.159	10286.360	40	left	3847.697	30.144	Change size	0.047	0.032	-
						0.007	TRUE	0.834				2571.590	10	right		120.577	Change size	0.047	0.032	-
7.31	0.0731	4.332	4.821	0.723	0.715	0.004	TRUE	0.715	4.181	11.726	160.410	1604.100	10	left	4008.107	165.684	Change size	0.020	0.025	-
						0.017	TRUE	0.715				6416.400	40	right		41.421	Change size	0.020	0.025	-
15.33	0.1533	6.274	6.982	1.047	1.035	0.015	TRUE	1.035	8.716	44.117	287.785	5755.700	20	left	4295.892	66.869	Change size	0.015	0.008	-
						0.008	TRUE	1.035				2877.850	10	right		133.739	Change size	0.015	0.008	-
-2.96	-0.0296	2.757	3.068	0.460	0.455	0.018	TRUE	0.455	(1.695)	4.838	163.447	6537.880	40	left	4459.339	25.868	Change size	0.041	0.010	slope change
						0.007	TRUE	0.455				2451.705	15	right		68.981	Change size	0.041	0.010	-
-0.61	-0.0061	1.252	1.393	0.209	0.207	0.015	TRUE	0.207	(0.349)	0.841	137.793	5511.720	40	left	4597.132	13.929	ECONOMICAL	0.024	0.004	-
						0.004	TRUE	0.207				1377.930	10	right		55.718	Change size	0.024	0.004	-
9.86	0.0986	5.032	5.599	0.840	0.830	0.009	TRUE	0.830	5.631	10.887	110.413	3312.390	30	left	4707.545	93.186	Change size	0.009	0.003	culvert
						0.003	TRUE	0.830				1104.130	10	right		279.558	Change size	0.009	0.003	slope change
-2.37	-0.0237	2.467	2.745	0.412	0.407	0.005	TRUE	0.407	(1.358)	1.335	56.332	1689.960	30	left	4763.877	89.547	Change size	0.049	0.002	culvert
						0.002	TRUE	0.407				563.320	10	right		268.642	Change size	0.049	0.002	-
9.93	0.0993	5.050	5.619	0.843	0.833	0.001	TRUE	0.833	5.671	2.641	26.595	531.900	20	left	4790.472	582.370	Change size	0.045	0.001	-
						0.001	TRUE	0.833				265.950	10	right		1164.740	Change size	0.045	0.001	-
8.52	0.0852	4.677	5.205	0.781	0.772	0.035	TRUE	0.772	4.870	24.974	293.125	13190.625	45	left	5083.597	21.752	Change size	0.043	0.008	culvert
						0.008	TRUE	0.772				2931.250	10	right		97.886	Change size	0.043	0.008	-
9.97	0.0997	5.060	5.630	0.845	0.835	0.008	TRUE	0.835	5.694	9.651	96.801	2904.030	30	left	5180.398	106.881	Change size	0.008	0.067	culvert
						0.003	TRUE	0.835				968.010	10	right		320.643	Change size	0.008	0.067	-
6.89	0.0689	4.206	4.681	0.702	0.694	0.005	TRUE	0.694	3.941	3.434	49.841	1993.640	40	left	5230.239	129.425	Change size	0.005	0.065	culvert
						0.001	TRUE	0.694				498.410	10	right		517.699	Change size	0.005	0.065	-
9.36	0.0936	4.902	5.455	0.818	0.809	0.041	TRUE	0.809	5.347	36.027	384.900	15396.000	40	left	5615.139	19.534	ECONOMICAL	0.041	0.063	culvert
						0.010	TRUE	0.809				3849.000	10	right		78.135	Change size	0.041	0.063	-
8.12	0.0812	4.566	5.081	0.762	0.753	0.029	TRUE	0.753	4.642	44.062	542.640	10852.800	20	left	6157.779	25.810	Change size	0.049	0.053	culvert
						0.015	TRUE	0.753				5426.400	10	right		51.620	Change size	0.049	0.053	-
10.76	0.1076	5.256	5.849	0.877	0.867	0.020	TRUE	0.867	6.141	22.865	212.501	0.000		Provided Same as min slope=9.94%	6370.280	-		0.020	0.038	-
						0.008	TRUE	0.867				0.000				-		0.020	0.038	-
9.99	0.0999	5.065	5.636	0.845	0.836	0.020	TRUE	0.836	5.705	8.949	89.577	0.000			6459.857	-		0.086	0.030	culvert
						0.008	TRUE	0.836				0.000				-		0.086	0.030	-
9.94	0.0994	5.052	5.622	0.843	0.834	0.020	TRUE	0.834	5.677	14.842	149.311	7465.550	50	left	6609.168	41.513	Change size	0.066	0.022	-
						0.008	TRUE	0.834				2986.220	20	right		103.783	Change size	0.066	0.022	-
6.95	0.0695	4.224	4.701	0.705	0.697	0.012	TRUE	0.697	3.976	6.200	89.209	4460.450	50	left	6698.377	58.099	Change size	0.046	0.014	-
						0.002	TRUE	0.697				892.090	10	right		290.495	Change size	0.046	0.014	-
2.21	0.0221	2.382	2.651	0.398	0.393	0.011	TRUE	0.393	1.266	2.175	98.424	3936.960	40	left	6796.801	37.118	Change size	0.034	0.012	-
						0.004	TRUE	0.393				1476.360	15	right		98.982	Change size	0.034	0.012	-
9.91	0.0991	5.044	5.613	0.842	0.832	0.013	TRUE	0.832	5.660	9.712	98.005	4900.250	50	left	6894.806	63.150	Change size	0.023	0.008	-
						0.004	TRUE	0.832				1470.075	15	right		210.500	Change size	0.023	0.008	-
9.84	0.0984	5.027	5.593	0.839	0.829	0.010	TRUE	0.829	5.620	9.272	94.230	3769.200	40	left	6989.036	81.809	Change size	0.010	0.004	-
						0.004	TRUE	0.829				1413.450	15	right		218.158	Change size	0.010	0.004	-

Qd/Qs for G-Drain(Using cumulative flow) left right		Qd/Qs for F-Drain(Using cumulative flow) left right			V _{dG}	Q _{d(Drain-G)}	Qd/Qs		Qd/Qs for G-Drain(Using cumulative flow) left right			V _{dF}	Q _{d(Drain-F)}	Qd/Qs		Qd/Qs for F-Drain(Using cumulative flow) left right	
14.888	24.592	15.060	24.878		2.970	0.458	97.476	13.696	22.624		3.360	0.504	107.183	15.060	24.877		
							77.981						85.747				
12.402	24.805	12.546	25.093		2.127	0.328	34.229	11.410	22.819		2.405	0.361	37.637	12.546	25.092		
							68.457						75.275				
17.168	34.336	17.367	34.735		1.963	0.303	31.588	15.794	31.588		2.220	0.333	34.734	17.367	34.734		
							63.176						69.467				
32.399	64.797	32.775	65.550		1.852	0.286	29.805	29.805	59.610		2.095	0.314	32.773	32.773	65.547		
							59.610						65.547				
1.792	10.574	1.813	10.697		2.739	0.423	22.009	1.648	9.728		3.099	0.465	24.200	1.812	10.697		
							110.043						121.002				
1.179	7.064	1.193	7.146		1.668	0.257	12.920	1.085	6.499		1.887	0.283	14.206	1.193	7.146		
							21.533						23.677				
2.448	19.240	2.477	19.463		3.172	0.490	77.924	2.252	17.700		3.588	0.538	85.685	2.477	19.462		
							116.886						128.527				
3.960	35.609	4.006	36.023		4.982	0.769	63.609	3.643	32.759		5.636	0.845	69.944	4.006	36.021		
							95.414						104.916				
3.153	40.709	3.190	41.182		3.740	0.577	149.801	2.901	37.450		4.231	0.635	164.720	3.190	41.180		
							37.450						41.180				
4.277	14.180	4.327	14.345		4.975	0.768	356.025	3.935	13.045		5.627	0.844	391.481	4.327	14.344		
							178.013						195.741				
2.459	8.700	2.488	8.801		2.829	0.436	184.389	2.262	8.004		3.200	0.480	202.752	2.487	8.801		
							92.195						101.376				
4.023	15.397	4.070	15.576		4.572	0.705	72.985	3.701	14.165		5.171	0.776	80.254	4.070	15.575		
							218.955						240.761				
4.622	17.951	4.675	18.160		4.985	0.769	157.965	4.252	16.514		5.639	0.846	173.696	4.675	18.159		
							78.983						86.848				
1.124	5.371	1.137	5.433		1.180	0.182	36.134	1.034	4.941		1.334	0.200	39.733	1.137	5.433		
							36.134						39.733				
4.082	21.950	4.129	22.205		4.162	0.642	99.167	3.755	20.193		4.707	0.706	109.043	4.129	22.204		
							99.167						109.043				
3.104	20.163	3.140	20.397		3.044	0.470	89.393	2.855	18.549		3.444	0.517	98.295	3.140	20.397		
							178.786						196.591				
4.464	31.322	4.516	31.686		4.239	0.654	107.863	4.106	28.815		4.794	0.719	118.605	4.515	31.684		
							215.726						237.210				
2.623	20.434	2.653	20.671		2.396	0.370	99.697	2.413	18.798		2.710	0.407	109.625	2.653	20.670		
							199.394						219.251				
3.768	31.628	3.812	31.996		3.359	0.518	186.906	3.467	29.097		3.799	0.570	205.519	3.812	31.994		
							186.906						205.519				
3.595	35.074	3.637	35.481		3.145	0.485	223.768	3.307	32.266		3.557	0.534	246.052	3.637	35.479		
							223.768						246.052				
6.309	70.866	6.383	71.689		5.438	0.839	21.731	5.804	65.193		6.151	0.923	23.895	6.382	71.686		
							65.193						71.686				
7.680	82.852	7.769	83.815		4.851	0.749	306.552	7.065	76.220		5.487	0.823	337.081	7.769	83.811		
							153.276						168.541				

Qd/Qs for G-Drain(Using cumulative flow) left right		Qd/Qs for F-Drain(Using cumulative flow) left right			V _{dG}	Q _{d(Drain-G)}	Qd/Qs		Qd/Qs for G-Drain(Using cumulative flow) left right		V _{dF}	Q _{d(Drain-F)}	Qd/Qs		Qd/Qs for F-Drain(Using cumulative flow) left right	
9.850	206.517	9.965	208.915		6.079	0.938	63.329	9.062	189.986		6.876	1.031	69.635	9.964	208.906	
							189.986						208.906			
9.792	16.406	9.906	16.597		5.178	0.799	26.099	9.008	15.093		5.857	0.879	28.698	9.905	16.596	
							43.499						47.831			
12.972	21.794	13.123	22.047		4.492	0.693	65.281	11.934	20.050		5.081	0.762	71.782	13.122	22.046	
							261.124						287.129			
17.572	26.132	17.776	26.435		4.972	0.767	27.731	16.165	24.040		5.624	0.844	30.493	17.775	26.434	
							110.925						121.972			
36.112	28.596	36.531	28.928		4.262	0.658	152.422	33.221	26.307		4.821	0.723	167.601	36.529	28.926	
							38.105						41.900			
66.869	133.739	67.646	135.292		6.172	0.952	61.517	61.517	123.033		6.981	1.047	67.643	67.643	135.286	
							123.033						135.286			
11.009	44.161	11.137	44.674		2.712	0.418	23.797	10.128	40.626		3.068	0.460	26.167	11.136	44.672	
							63.460						69.779			
8.701	55.718	8.802	56.365		1.231	0.190	12.814	8.004	51.258		1.393	0.209	14.091	8.801	56.362	
							51.258						56.362			
93.186	279.558	94.268	282.805		4.950	0.764	85.727	85.727	257.181		5.599	0.840	94.264	94.264	282.793	
							257.181						282.793			
8.262	182.487	8.358	184.606		2.427	0.374	82.379	7.601	167.880		2.745	0.412	90.583	8.358	184.598	
							247.138						271.750			
18.631	1164.740	18.847	1178.265		4.967	0.766	535.753	17.139	1071.506		5.619	0.843	589.107	18.846	1178.215	
							1071.506						1178.215			
17.828	97.886	18.035	99.023		4.601	0.710	20.011	16.401	90.051		5.205	0.781	22.004	18.034	99.019	
							90.051						99.019			
106.881	12.439	108.122	12.584		4.977	0.768	98.326	98.326	11.443		5.630	0.845	108.118	108.118	12.583	
							294.977						324.353			
129.425	10.758	130.928	10.883		4.138	0.638	119.065	119.065	9.897		4.680	0.702	130.922	130.922	10.883	
							476.258						523.688			
19.534	12.805	19.761	12.954		4.823	0.744	17.970	17.970	11.780		5.455	0.818	19.760	19.760	12.953	
							71.880						79.039			
15.291	14.264	15.469	14.430		4.492	0.693	23.744	14.067	13.123		5.081	0.762	26.109	15.468	14.430	
							47.488						52.217			
43.192	22.691	43.693	22.954		5.171	0.798	39.734	39.734	20.874		5.849	0.877	43.691	43.691	22.953	
							99.335						109.228			
9.710	27.680	9.823	28.002		4.982	0.769	38.286	8.933	25.465		5.636	0.845	42.099	9.822	28.001	
							95.715						105.247			
12.633	37.620	12.780	38.057		4.970	0.767	38.190	11.622	34.608		5.622	0.843	41.993	12.779	38.055	
							95.475						104.984			
15.184	49.343	15.361	49.916		4.156	0.641	53.448	13.969	45.393		4.701	0.705	58.771	15.360	49.914	
							267.241						293.855			
11.592	33.518	11.727	33.907		2.343	0.362	34.147	10.664	30.835		2.651	0.398	37.548	11.726	33.906	
							91.059						100.128			
35.694	107.317	36.109	108.563		4.962	0.766	58.095	32.837	98.726		5.613	0.842	63.881	36.107	108.558	
							193.650						212.935			
81.809	218.158	82.759	220.691		4.945	0.763	75.261	75.261	200.695		5.593	0.839	82.756	82.756	220.682	
							200.695						220.682			

For culvert

Culvert Locations	Catchment Area(m ²)	Discharge from hill	Drain discharge	Total culvert discharge	Remark
6450 km	45781.250	0.210	0.086	0.296	
5800km	356093.750	1.637	0.049	1.686	
5610km	34062.500	0.157	0.041	0.198	
5220km	1718.750	0.008	0.005	0.013	Pani Ghata
5100km	2031.250	0.009	0.067	0.076	Pani Ghata
4873.629km	87500.000	0.402	0.008	0.410	
4720km	87615.193	0.403	0.049	0.452	consecutive
4600km	14843.750	0.068	0.003	0.071	outlet
3270km	781.250	0.004	0.053	0.057	
2790km	781.250	0.004	0.010	0.013	
1200km	312.500	0.001	0.059	0.060	
700km	12500.000	0.057	0.083	0.141	causeway

Slope(decimal)	Slope of culvert	D(assume)	Thita(θ)	A	P	R	Qgoal	Theta(deg)	Y(water depth)	CHECK(y<D)
0.050	0.050	0.900	1.769	0.080	0.796	0.100	0.296	101.358	0.165	DESIGN OK
0.050	0.050	0.900	2.941	0.278	1.324	0.210	1.686	168.517	0.405	DESIGN OK
0.050	0.050	0.900	1.593	0.060	0.717	0.084	0.198	91.288	0.135	DESIGN OK
0.050	0.050	0.900	0.824	0.009	0.371	0.025	0.013	47.196	0.038	DESIGN OK
0.050	0.050	0.900	1.256	0.031	0.565	0.055	0.076	71.943	0.086	DESIGN OK
0.050	0.050	0.900	1.928	0.100	0.868	0.116	0.410	110.488	0.193	DESIGN OK
0.050	0.050	0.900	1.980	0.108	0.891	0.121	0.452	113.439	0.203	DESIGN OK
0.050	0.050	0.900	1.234	0.029	0.555	0.053	0.071	70.696	0.083	DESIGN OK
0.050	0.050	0.900	1.166	0.025	0.525	0.048	0.057	66.821	0.074	DESIGN OK
0.050	0.050	0.900	0.826	0.009	0.372	0.025	0.013	47.321	0.038	DESIGN OK
0.050	0.050	0.900	1.185	0.026	0.533	0.049	0.060	67.876	0.077	DESIGN OK

VELOCITY TABLE

Drain- TYPE G	B	S	y		A		P		R		Qset		Qgoal		Velocity(V)		Check V=0.6-5 m/s	
	0.440	3.550	0.045	0.033	0.020	0.015	0.536	0.510	0.038	0.029	0.033	0.020	0.033	0.020	1.640	1.369	TRUE	TRUE
	0.440	3.550	0.045	0.033	0.020	0.015	0.536	0.510	0.038	0.029	0.033	0.020	0.033	0.020	1.640	1.369	TRUE	TRUE
	0.440	1.820	0.051	0.033	0.023	0.015	0.547	0.510	0.042	0.029	0.029	0.014	0.029	0.014	1.253	0.977	TRUE	TRUE
	0.440	1.820	0.051	0.033	0.023	0.015	0.547	0.510	0.042	0.029	0.029	0.014	0.029	0.014	1.253	0.977	TRUE	TRUE
	0.440	1.550	0.041	0.027	0.019	0.012	0.528	0.497	0.035	0.024	0.019	0.010	0.019	0.010	1.030	0.800	TRUE	TRUE
	0.440	1.550	0.041	0.027	0.019	0.012	0.528	0.497	0.035	0.024	0.019	0.010	0.019	0.010	1.030	0.800	TRUE	TRUE
	0.440	1.380	0.028	0.018	0.012	0.008	0.499	0.478	0.025	0.017	0.010	0.005	0.010	0.005	0.771	0.594	TRUE	FALSE
	0.440	1.380	0.028	0.018	0.012	0.008	0.499	0.478	0.025	0.017	0.010	0.005	0.010	0.005	0.771	0.594	TRUE	FALSE
	0.440	(3.020)	0.178	0.056	0.086	0.025	0.816	0.559	0.105	0.046	0.256	0.043	0.256	0.043	2.982	1.706	TRUE	TRUE
	0.440	(3.020)	0.178	0.056	0.086	0.025	0.816	0.559	0.105	0.046	0.256	0.043	0.256	0.043	2.982	1.706	TRUE	TRUE
	0.440	1.120	0.234	0.073	0.117	0.033	0.936	0.594	0.125	0.056	0.237	0.040	0.237	0.040	2.032	1.192	TRUE	TRUE
	0.440	1.120	0.234	0.073	0.117	0.033	0.936	0.594	0.125	0.056	0.237	0.040	0.237	0.040	2.032	1.192	TRUE	TRUE
	0.440	4.050	0.144	0.038	0.069	0.017	0.746	0.522	0.092	0.033	0.217	0.028	0.217	0.028	3.159	1.598	TRUE	TRUE
	0.440	4.050	0.144	0.038	0.069	0.017	0.746	0.522	0.092	0.033	0.217	0.028	0.217	0.028	3.159	1.598	TRUE	TRUE
	0.440	9.990	0.105	0.026	0.049	0.012	0.663	0.496	0.074	0.024	0.211	0.023	0.211	0.023	4.291	2.003	TRUE	TRUE
	0.440	9.990	0.105	0.026	0.049	0.012	0.663	0.496	0.074	0.024	0.211	0.023	0.211	0.023	4.291	2.003	TRUE	TRUE
	0.440	5.630	0.122	0.024	0.058	0.011	0.699	0.491	0.082	0.022	0.199	0.015	0.199	0.015	3.455	1.430	TRUE	TRUE
	0.440	5.630	0.122	0.024	0.058	0.011	0.699	0.491	0.082	0.022	0.199	0.015	0.199	0.015	3.455	1.430	TRUE	TRUE
	0.440	9.960	0.100	0.047	0.047	0.021	0.652	0.539	0.072	0.039	0.195	0.059	0.195	0.059	4.182	2.795	TRUE	TRUE
	0.440	9.960	0.100	0.047	0.047	0.021	0.652	0.539	0.072	0.039	0.195	0.059	0.195	0.059	4.182	2.795	TRUE	TRUE
	0.440	3.220	0.144	0.064	0.069	0.029	0.745	0.575	0.092	0.050	0.193	0.055	0.193	0.055	2.814	1.883	TRUE	TRUE
	0.440	3.220	0.144	0.064	0.069	0.029	0.745	0.575	0.092	0.050	0.193	0.055	0.193	0.055	2.814	1.883	TRUE	TRUE
	0.440	8.410	0.104	0.044	0.049	0.020	0.661	0.534	0.074	0.037	0.191	0.050	0.191	0.050	3.917	2.495	TRUE	TRUE
	0.440	8.410	0.104	0.044	0.049	0.020	0.661	0.534	0.074	0.037	0.191	0.050	0.191	0.050	3.917	2.495	TRUE	TRUE
	0.440	10.000	0.095	0.040	0.044	0.018	0.642	0.525	0.069	0.034	0.181	0.047	0.181	0.047	4.089	2.575	TRUE	TRUE
	0.440	10.000	0.095	0.040	0.044	0.018	0.642	0.525	0.069	0.034	0.181	0.047	0.181	0.047	4.089	2.575	TRUE	TRUE
	0.440	0.560	0.242	0.087	0.121	0.040	0.952	0.623	0.127	0.064	0.176	0.037	0.176	0.037	1.455	0.922	TRUE	TRUE
	0.440	0.560	0.242	0.087	0.121	0.040	0.952	0.623	0.127	0.064	0.176	0.037	0.176	0.037	1.455	0.922	TRUE	TRUE
	0.440	6.970	0.103	0.035	0.048	0.016	0.659	0.515	0.073	0.031	0.171	0.032	0.171	0.032	3.550	1.999	TRUE	TRUE
	0.440	6.970	0.103	0.035	0.048	0.016	0.659	0.515	0.073	0.031	0.171	0.032	0.171	0.032	3.550	1.999	TRUE	TRUE
	0.440	3.730	0.124	0.037	0.058	0.017	0.702	0.519	0.083	0.032	0.165	0.025	0.165	0.025	2.826	1.508	TRUE	TRUE
	0.440	3.730	0.124	0.037	0.058	0.017	0.702	0.519	0.083	0.032	0.165	0.025	0.165	0.025	2.826	1.508	TRUE	TRUE
	0.440	7.230	0.098	0.028	0.045	0.013	0.647	0.500	0.070	0.025	0.159	0.023	0.159	0.023	3.515	1.787	TRUE	TRUE
	0.440	7.230	0.098	0.028	0.045	0.013	0.647	0.500	0.070	0.025	0.159	0.023	0.159	0.023	3.515	1.787	TRUE	TRUE
	0.440	2.310	0.138	0.037	0.066	0.017	0.732	0.518	0.089	0.032	0.153	0.020	0.153	0.020	2.338	1.181	TRUE	TRUE
	0.440	2.310	0.138	0.037	0.066	0.017	0.732	0.518	0.089	0.032	0.153	0.020	0.153	0.020	2.338	1.181	TRUE	TRUE
	0.440	4.540	0.109	0.028	0.051	0.013	0.671	0.500	0.076	0.025	0.149	0.018	0.149	0.018	2.938	1.411	TRUE	TRUE
	0.440	4.540	0.109	0.028	0.051	0.013	0.671	0.500	0.076	0.025	0.149	0.018	0.149	0.018	2.938	1.411	TRUE	TRUE

Drain- TYPE G	B	S	y		A		P		R		Qset		Qgoal		Velocity(V)		Check V=0.6-5 m/s	
	0.440	3.980	0.112	0.026	0.053	0.012	0.678	0.496	0.078	0.024	0.147	0.015	0.147	0.015	2.791	1.271	TRUE	TRUE
	0.440	3.980	0.112	0.026	0.053	0.012	0.678	0.496	0.078	0.024	0.147	0.015	0.147	0.015	2.791	1.271	TRUE	TRUE
	0.440	11.900	0.078	0.017	0.036	0.008	0.605	0.476	0.059	0.016	0.145	0.013	0.145	0.013	4.032	1.686	TRUE	TRUE
	0.440	11.900	0.078	0.017	0.036	0.008	0.605	0.476	0.059	0.016	0.145	0.013	0.145	0.013	4.032	1.686	TRUE	TRUE
	0.440	9.470	0.069	0.016	0.031	0.007	0.586	0.473	0.054	0.015	0.106	0.010	0.106	0.010	3.369	1.417	TRUE	TRUE
	0.440	9.470	0.069	0.016	0.031	0.007	0.586	0.473	0.054	0.015	0.106	0.010	0.106	0.010	3.369	1.417	TRUE	TRUE
	0.440	14.870	0.059	0.009	0.027	0.004	0.564	0.459	0.047	0.009	0.104	0.005	0.104	0.005	3.879	1.247	TRUE	TRUE
	0.440	14.870	0.059	0.009	0.027	0.004	0.564	0.459	0.047	0.009	0.104	0.005	0.104	0.005	3.879	1.247	TRUE	TRUE
	0.440	10.790	0.059	0.043	0.027	0.019	0.565	0.530	0.047	0.036	0.089	0.053	0.089	0.053	3.311	2.763	TRUE	TRUE
	0.440	10.790	0.059	0.043	0.027	0.019	0.565	0.530	0.047	0.036	0.089	0.053	0.089	0.053	3.311	2.763	TRUE	TRUE
	0.440	8.120	0.049	0.036	0.022	0.016	0.544	0.515	0.041	0.031	0.058	0.035	0.058	0.035	2.605	2.163	TRUE	TRUE
	0.440	8.120	0.049	0.036	0.022	0.016	0.544	0.515	0.041	0.031	0.058	0.035	0.058	0.035	2.605	2.163	TRUE	TRUE
	0.440	9.950	0.041	0.032	0.018	0.014	0.526	0.507	0.035	0.028	0.047	0.032	0.047	0.032	2.588	2.241	TRUE	TRUE
	0.440	9.950	0.041	0.032	0.018	0.014	0.526	0.507	0.035	0.028	0.047	0.032	0.047	0.032	2.588	2.241	TRUE	TRUE
	0.440	7.310	0.026	0.030	0.012	0.013	0.495	0.504	0.023	0.027	0.020	0.025	0.020	0.025	1.704	1.858	TRUE	TRUE
	0.440	7.310	0.026	0.030	0.012	0.013	0.495	0.504	0.023	0.027	0.020	0.025	0.020	0.025	1.704	1.858	TRUE	TRUE
	0.440	15.330	0.018	0.012	0.008	0.005	0.478	0.465	0.017	0.011	0.015	0.008	0.015	0.008	1.957	1.500	TRUE	TRUE
	0.440	15.330	0.018	0.012	0.008	0.005	0.478	0.465	0.017	0.011	0.015	0.008	0.015	0.008	1.957	1.500	TRUE	TRUE
	0.440	(2.960)	0.055	0.023	0.025	0.010	0.556	0.489	0.045	0.021	0.041	0.010	0.041	0.010	1.665	1.006	TRUE	TRUE
	0.440	(2.960)	0.055	0.023	0.025	0.010	0.556	0.489	0.045	0.021	0.041	0.010	0.041	0.010	1.665	1.006	TRUE	TRUE
	0.440	(0.610)	0.064	0.020	0.029	0.009	0.575	0.482	0.050	0.018	0.024	0.004	0.024	0.004	0.820	0.418	TRUE	FALSE
	0.440	(0.610)	0.064	0.020	0.029	0.009	0.575	0.482	0.050	0.018	0.024	0.004	0.024	0.004	0.820	0.418	TRUE	FALSE
	0.440	9.860	0.015	0.007	0.006	0.003	0.471	0.456	0.014	0.007	0.009	0.003	0.009	0.003	1.382	0.902	TRUE	TRUE
	0.440	9.860	0.015	0.007	0.006	0.003	0.471	0.456	0.014	0.007	0.009	0.003	0.009	0.003	1.382	0.902	TRUE	TRUE
	0.440	(2.370)	0.066	0.010	0.030	0.004	0.579	0.460	0.052	0.009	0.049	0.002	0.049	0.002	1.644	0.523	TRUE	FALSE
	0.440	(2.370)	0.066	0.010	0.030	0.004	0.579	0.460	0.052	0.009	0.049	0.002	0.049	0.002	1.644	0.523	TRUE	FALSE
	0.440	9.930	0.039	0.003	0.018	0.001	0.523	0.447	0.034	0.003	0.045	0.001	0.045	0.001	2.532	0.516	TRUE	FALSE
	0.440	9.930	0.039	0.003	0.018	0.001	0.523	0.447	0.034	0.003	0.045	0.001	0.045	0.001	2.532	0.516	TRUE	FALSE
	0.440	8.520	0.040	0.014	0.018	0.006	0.525	0.470	0.035	0.013	0.043	0.008	0.043	0.008	2.383	1.261	TRUE	TRUE
	0.440	8.520	0.040	0.014	0.018	0.006	0.525	0.470	0.035	0.013	0.043	0.008	0.043	0.008	2.383	1.261	TRUE	TRUE
	0.440	9.970	0.013	0.051	0.006	0.023	0.468	0.547	0.013	0.042	0.008	0.067	0.008	0.067	1.319	2.929	TRUE	TRUE
	0.440	9.970	0.013	0.051	0.006	0.023	0.468	0.547	0.013	0.042	0.008	0.067	0.008	0.067	1.319	2.929	TRUE	TRUE
	0.440	6.890	0.012	0.055	0.005	0.025	0.465	0.558	0.011	0.045	0.005	0.065	0.005	0.065	1.018	2.561	TRUE	TRUE
	0.440	6.890	0.012	0.055	0.005	0.025	0.465	0.558	0.011	0.045	0.005	0.065	0.005	0.065	1.018	2.561	TRUE	TRUE
	0.440	9.360	0.038	0.050	0.017	0.022	0.521	0.545	0.033	0.041	0.041	0.063	0.041	0.063	2.417	2.809	TRUE	TRUE
	0.440	9.360	0.038	0.050	0.017	0.022	0.521	0.545	0.033	0.041	0.041	0.063	0.041	0.063	2.417	2.809	TRUE	TRUE
	0.440	8.120	0.044	0.046	0.020	0.021	0.534	0.538	0.038	0.039	0.049	0.053	0.049	0.053	2.457	2.519	TRUE	TRUE
	0.440	8.120	0.044	0.046	0.020	0.021	0.534	0.538	0.038	0.039	0.049	0.053	0.049	0.053	2.457	2.519	TRUE	TRUE
	0.440	10.760	0.023	0.035	0.010	0.016	0.489	0.514	0.021	0.030	0.020	0.038	0.020	0.038	1.934	2.454	TRUE	TRUE
	0.440	10.760	0.023	0.035	0.010	0.016	0.489	0.514	0.021	0.030	0.020	0.038	0.020	0.038	1.934	2.454	TRUE	TRUE

Drain- TYPE G	B	S	y		A		P		R		Qset		Qgoal		Velocity(V)		Check V=0.6-5 m/s	
	0.440	9.990	0.059	0.031	0.027	0.014	0.565	0.505	0.048	0.027	0.086	0.030	0.086	0.030	3.195	2.199	TRUE	TRUE
	0.440	9.990	0.059	0.031	0.027	0.014	0.565	0.505	0.048	0.027	0.086	0.030	0.086	0.030	3.195	2.199	TRUE	TRUE
	0.440	9.940	0.050	0.025	0.023	0.011	0.546	0.494	0.042	0.023	0.066	0.022	0.066	0.022	2.909	1.957	TRUE	TRUE
	0.440	9.990	0.050	0.025	0.023	0.011	0.546	0.494	0.041	0.023	0.066	0.022	0.066	0.022	2.913	1.960	TRUE	TRUE
	0.440	6.950	0.045	0.021	0.020	0.010	0.535	0.485	0.038	0.020	0.046	0.014	0.046	0.014	2.279	1.478	TRUE	TRUE
	0.440	6.950	0.045	0.021	0.020	0.010	0.535	0.485	0.038	0.020	0.046	0.014	0.046	0.014	2.279	1.478	TRUE	TRUE
	0.440	2.210	0.053	0.027	0.024	0.012	0.552	0.498	0.043	0.024	0.034	0.012	0.034	0.012	1.413	0.963	TRUE	TRUE
	0.440	2.210	0.053	0.027	0.024	0.012	0.552	0.498	0.043	0.024	0.034	0.012	0.034	0.012	1.413	0.963	TRUE	TRUE
	0.440	9.910	0.026	0.013	0.012	0.006	0.495	0.468	0.024	0.013	0.023	0.008	0.023	0.008	1.993	1.313	TRUE	TRUE
	0.440	9.910	0.026	0.013	0.012	0.006	0.495	0.468	0.024	0.013	0.023	0.008	0.023	0.008	1.993	1.313	TRUE	TRUE
	0.440	9.840	0.016	0.009	0.007	0.004	0.473	0.458	0.015	0.008	0.010	0.004	0.010	0.004	1.452	0.993	TRUE	TRUE
	0.440	9.840	0.016	0.009	0.007	0.004	0.473	0.458	0.015	0.008	0.010	0.004	0.010	0.004	1.452	0.993	TRUE	TRUE

Drain- TYPE F	S	y		A		P		R		Qset		Qgoal		Velocity(V)		Check V=0.6-5 m/s	
	3.550	0.111	0.092	0.020	0.013	0.481	0.398	0.041	0.034	0.033	0.020	0.033	0.020	1.713	1.511	TRUE	TRUE
	3.550	0.111	0.092	0.020	0.013	0.481	0.398	0.041	0.034	0.033	0.020	0.033	0.020	1.713	1.511	TRUE	TRUE
	1.820	0.118	0.091	0.022	0.013	0.515	0.397	0.044	0.034	0.029	0.014	0.029	0.014	1.284	1.079	TRUE	TRUE
	1.820	0.118	0.091	0.022	0.013	0.515	0.397	0.044	0.034	0.029	0.014	0.029	0.014	1.284	1.079	TRUE	TRUE
	1.550	0.105	0.081	0.018	0.010	0.456	0.352	0.039	0.030	0.019	0.010	0.019	0.010	1.092	0.918	TRUE	TRUE
	1.550	0.105	0.081	0.018	0.010	0.456	0.352	0.039	0.030	0.019	0.010	0.019	0.010	1.092	0.918	TRUE	TRUE
	1.380	0.083	0.064	0.011	0.006	0.359	0.277	0.030	0.023	0.010	0.005	0.010	0.005	0.879	0.739	TRUE	TRUE
	1.380	0.083	0.064	0.011	0.006	0.359	0.277	0.030	0.023	0.010	0.005	0.010	0.005	0.879	0.739	TRUE	TRUE
	(3.020)	0.244	0.126	0.096	0.025	1.064	0.547	0.090	0.046	0.256	0.043	0.256	0.043	2.682	1.721	TRUE	TRUE
	(3.020)	0.244	0.126	0.096	0.025	1.064	0.547	0.090	0.046	0.256	0.043	0.256	0.043	2.682	1.721	TRUE	TRUE
	1.120	0.250	0.146	0.100	0.034	1.088	0.636	0.092	0.054	0.166	0.040	0.237	0.040	1.658	1.159	TRUE	TRUE
	1.120	0.250	0.146	0.100	0.034	1.088	0.636	0.092	0.054	0.166	0.040	0.237	0.040	1.658	1.159	TRUE	TRUE
	4.050	0.217	0.100	0.076	0.016	0.946	0.437	0.080	0.037	0.217	0.028	0.217	0.028	2.873	1.716	TRUE	TRUE
	4.050	0.217	0.100	0.076	0.016	0.946	0.437	0.080	0.037	0.217	0.028	0.217	0.028	2.873	1.716	TRUE	TRUE
	9.990	0.182	0.080	0.053	0.010	0.790	0.347	0.067	0.029	0.211	0.023	0.211	0.023	4.001	2.310	TRUE	TRUE
	9.990	0.182	0.080	0.053	0.010	0.790	0.347	0.067	0.029	0.211	0.023	0.211	0.023	4.001	2.310	TRUE	TRUE
	5.630	0.198	0.076	0.063	0.009	0.861	0.330	0.073	0.028	0.199	0.015	0.199	0.015	3.179	1.677	TRUE	TRUE
	5.630	0.198	0.076	0.063	0.009	0.861	0.330	0.073	0.028	0.199	0.015	0.199	0.015	3.179	1.677	TRUE	TRUE
	9.960	0.176	0.113	0.050	0.020	0.768	0.490	0.065	0.041	0.195	0.059	0.195	0.059	3.918	2.904	TRUE	TRUE
	9.960	0.176	0.113	0.050	0.020	0.768	0.490	0.065	0.041	0.195	0.059	0.195	0.059	3.918	2.904	TRUE	TRUE
	3.220	0.217	0.135	0.075	0.029	0.945	0.588	0.080	0.050	0.193	0.055	0.193	0.055	2.559	1.866	TRUE	TRUE
	3.220	0.217	0.135	0.075	0.029	0.945	0.588	0.080	0.050	0.193	0.055	0.193	0.055	2.559	1.866	TRUE	TRUE
	8.410	0.180	0.109	0.052	0.019	0.786	0.475	0.066	0.040	0.191	0.050	0.191	0.050	3.656	2.614	TRUE	TRUE
	8.410	0.180	0.109	0.052	0.019	0.786	0.475	0.066	0.040	0.191	0.050	0.191	0.050	3.656	2.614	TRUE	TRUE
	10.000	0.171	0.103	0.047	0.017	0.746	0.448	0.063	0.038	0.181	0.047	0.181	0.047	3.851	2.743	TRUE	TRUE
	10.000	0.171	0.103	0.047	0.017	0.746	0.448	0.063	0.038	0.181	0.047	0.181	0.047	3.851	2.743	TRUE	TRUE
	0.560	0.250	0.162	0.100	0.042	1.088	0.705	0.092	0.060	0.117	0.037	0.176	0.037	1.172	0.878	TRUE	TRUE
	0.560	0.250	0.162	0.100	0.042	1.088	0.705	0.092	0.060	0.117	0.037	0.176	0.037	1.172	0.878	TRUE	TRUE
	6.970	0.180	0.096	0.052	0.015	0.781	0.416	0.066	0.035	0.171	0.032	0.171	0.032	3.316	2.178	TRUE	TRUE
	6.970	0.180	0.096	0.052	0.015	0.781	0.416	0.066	0.035	0.171	0.032	0.171	0.032	3.316	2.178	TRUE	TRUE
	3.730	0.199	0.099	0.063	0.016	0.866	0.429	0.073	0.036	0.165	0.025	0.165	0.025	2.598	1.627	TRUE	TRUE
	3.730	0.199	0.099	0.063	0.016	0.866	0.429	0.073	0.036	0.165	0.025	0.165	0.025	2.598	1.627	TRUE	TRUE
	7.230	0.174	0.084	0.048	0.011	0.756	0.364	0.064	0.031	0.159	0.023	0.159	0.023	3.303	2.029	TRUE	TRUE
	7.230	0.174	0.084	0.048	0.011	0.756	0.364	0.064	0.031	0.159	0.023	0.159	0.023	3.303	2.029	TRUE	TRUE
	2.310	0.212	0.098	0.072	0.015	0.922	0.427	0.078	0.036	0.153	0.020	0.153	0.020	2.132	1.276	TRUE	TRUE
	2.310	0.212	0.098	0.072	0.015	0.922	0.427	0.078	0.036	0.153	0.020	0.153	0.020	2.132	1.276	TRUE	TRUE
	4.540	0.185	0.083	0.055	0.011	0.805	0.363	0.068	0.031	0.149	0.018	0.149	0.018	2.731	1.604	TRUE	TRUE
	4.540	0.185	0.083	0.055	0.011	0.805	0.363	0.068	0.031	0.149	0.018	0.149	0.018	2.731	1.604	TRUE	TRUE

Drain- TYPE F	S	y		A		P		R		Qset		Qgoal		Velocity(V)		Check V=0.6-5 m/s	
	3.980	0.188	0.080	0.057	0.010	0.820	0.349	0.069	0.029	0.147	0.015	0.147	0.015	2.587	1.464	TRUE	TRUE
	3.980	0.188	0.080	0.057	0.010	0.820	0.349	0.069	0.029	0.147	0.015	0.147	0.015	2.587	1.464	TRUE	TRUE
	11.900	0.152	0.062	0.037	0.006	0.664	0.268	0.056	0.023	0.145	0.013	0.145	0.013	3.886	2.123	TRUE	TRUE
	11.900	0.152	0.062	0.037	0.006	0.664	0.268	0.056	0.023	0.145	0.013	0.145	0.013	3.886	2.123	TRUE	TRUE
	9.470	0.142	0.058	0.032	0.005	0.616	0.253	0.052	0.021	0.106	0.010	0.106	0.010	3.301	1.821	TRUE	TRUE
	9.470	0.142	0.058	0.032	0.005	0.616	0.253	0.052	0.021	0.106	0.010	0.106	0.010	3.301	1.821	TRUE	TRUE
	14.870	0.129	0.041	0.027	0.003	0.562	0.179	0.047	0.015	0.104	0.005	0.104	0.005	3.886	1.816	TRUE	TRUE
	14.870	0.129	0.041	0.027	0.003	0.562	0.179	0.047	0.015	0.104	0.005	0.104	0.005	3.886	1.816	TRUE	TRUE
	10.790	0.129	0.107	0.027	0.018	0.563	0.464	0.048	0.039	0.089	0.053	0.089	0.053	3.316	2.914	TRUE	TRUE
	10.790	0.129	0.107	0.027	0.018	0.563	0.464	0.048	0.039	0.089	0.053	0.089	0.053	3.316	2.914	TRUE	TRUE
	8.120	0.116	0.096	0.022	0.015	0.506	0.417	0.043	0.035	0.058	0.035	0.058	0.035	2.681	2.355	TRUE	TRUE
	8.120	0.116	0.096	0.022	0.015	0.506	0.417	0.043	0.035	0.058	0.035	0.058	0.035	2.681	2.355	TRUE	TRUE
	9.950	0.104	0.089	0.017	0.013	0.452	0.389	0.038	0.033	0.047	0.032	0.047	0.032	2.751	2.491	TRUE	TRUE
	9.950	0.104	0.089	0.017	0.013	0.452	0.389	0.038	0.033	0.047	0.032	0.047	0.032	2.751	2.491	TRUE	TRUE
	7.310	0.079	0.087	0.010	0.012	0.345	0.377	0.029	0.032	0.020	0.025	0.020	0.025	1.969	2.088	TRUE	TRUE
	7.310	0.079	0.087	0.010	0.012	0.345	0.377	0.029	0.032	0.020	0.025	0.020	0.025	1.969	2.088	TRUE	TRUE
	15.330	0.063	0.049	0.006	0.004	0.274	0.211	0.023	0.018	0.015	0.008	0.015	0.008	2.445	2.056	TRUE	TRUE
	15.330	0.063	0.049	0.006	0.004	0.274	0.211	0.023	0.018	0.015	0.008	0.015	0.008	2.445	2.056	TRUE	TRUE
	(2.960)	0.124	0.074	0.025	0.009	0.539	0.320	0.045	0.027	0.041	0.010	0.041	0.010	1.686	1.192	TRUE	TRUE
	(2.960)	0.124	0.074	0.025	0.009	0.539	0.320	0.045	0.027	0.041	0.010	0.041	0.010	1.686	1.192	TRUE	TRUE
	(0.610)	0.135	0.067	0.029	0.007	0.588	0.293	0.050	0.025	0.024	0.004	0.024	0.004	0.812	0.510	TRUE	FALSE
	(0.610)	0.135	0.067	0.029	0.007	0.588	0.293	0.050	0.025	0.024	0.004	0.024	0.004	0.812	0.510	TRUE	FALSE
	9.860	0.056	0.037	0.005	0.002	0.242	0.160	0.020	0.014	0.009	0.003	0.009	0.003	1.805	1.371	TRUE	TRUE
	9.860	0.056	0.037	0.005	0.002	0.242	0.160	0.020	0.014	0.009	0.003	0.009	0.003	1.805	1.371	TRUE	TRUE
	(2.370)	0.138	0.043	0.030	0.003	0.600	0.188	0.051	0.016	0.049	0.002	0.049	0.002	1.621	0.748	TRUE	TRUE
	(2.370)	0.138	0.043	0.030	0.003	0.600	0.188	0.051	0.016	0.049	0.002	0.049	0.002	1.621	0.748	TRUE	TRUE
	9.930	0.102	0.022	0.017	0.001	0.442	0.094	0.037	0.008	0.045	0.001	0.045	0.001	2.708	0.963	TRUE	TRUE
	9.930	0.102	0.022	0.017	0.001	0.442	0.094	0.037	0.008	0.045	0.001	0.045	0.001	2.708	0.963	TRUE	TRUE
	8.520	0.103	0.055	0.017	0.005	0.450	0.237	0.038	0.020	0.043	0.008	0.043	0.008	2.536	1.657	TRUE	TRUE
	8.520	0.103	0.055	0.017	0.005	0.450	0.237	0.038	0.020	0.043	0.008	0.043	0.008	2.536	1.657	TRUE	TRUE
	9.970	0.053	0.118	0.004	0.022	0.230	0.515	0.019	0.043	0.008	0.067	0.008	0.067	1.753	3.002	TRUE	TRUE
	9.970	0.053	0.118	0.004	0.022	0.230	0.515	0.019	0.043	0.008	0.067	0.008	0.067	1.753	3.002	TRUE	TRUE
	6.890	0.049	0.125	0.004	0.025	0.214	0.543	0.018	0.046	0.005	0.065	0.005	0.065	1.390	2.588	TRUE	TRUE
	6.890	0.049	0.125	0.004	0.025	0.214	0.543	0.018	0.046	0.005	0.065	0.005	0.065	1.390	2.588	TRUE	TRUE
	9.360	0.100	0.117	0.016	0.022	0.434	0.509	0.037	0.043	0.041	0.063	0.041	0.063	2.598	2.888	TRUE	TRUE
	9.360	0.100	0.117	0.016	0.022	0.434	0.509	0.037	0.043	0.041	0.063	0.041	0.063	2.598	2.888	TRUE	TRUE
	8.120	0.109	0.112	0.019	0.020	0.476	0.489	0.040	0.041	0.049	0.053	0.049	0.053	2.573	2.618	TRUE	TRUE
	8.120	0.109	0.112	0.019	0.020	0.476	0.489	0.040	0.041	0.049	0.053	0.049	0.053	2.573	2.618	TRUE	TRUE
	10.760	0.074	0.094	0.009	0.014	0.323	0.411	0.027	0.035	0.020	0.038	0.020	0.038	2.285	2.684	TRUE	TRUE
	10.760	0.074	0.094	0.009	0.014	0.323	0.411	0.027	0.035	0.020	0.038	0.020	0.038	2.285	2.684	TRUE	TRUE
	9.990	0.130	0.088	0.027	0.012	0.565	0.381	0.048	0.032	0.086	0.030	0.086	0.030	3.197	2.460	TRUE	TRUE

Drain- TYPE F	S	y		A		P		R		Qset		Qgoal		Velocity(V)		Check V=0.6-5 m/s	
	9.990	0.130	0.088	0.027	0.012	0.565	0.381	0.048	0.032	0.086	0.030	0.086	0.030	3.197	2.460	TRUE	TRUE
	9.940	0.118	0.078	0.022	0.010	0.512	0.340	0.043	0.029	0.066	0.022	0.066	0.022	2.986	2.273	TRUE	TRUE
	9.990	0.117	0.078	0.022	0.010	0.511	0.339	0.043	0.029	0.066	0.022	0.066	0.022	2.992	2.277	TRUE	TRUE
	6.950	0.110	0.071	0.019	0.008	0.477	0.307	0.040	0.026	0.046	0.014	0.046	0.014	2.385	1.776	TRUE	TRUE
	6.950	0.110	0.071	0.019	0.008	0.477	0.307	0.040	0.026	0.046	0.014	0.046	0.014	2.385	1.776	TRUE	TRUE
	2.210	0.121	0.082	0.024	0.011	0.528	0.355	0.045	0.030	0.034	0.012	0.034	0.012	1.439	1.103	TRUE	TRUE
	2.210	0.121	0.082	0.024	0.011	0.528	0.355	0.045	0.030	0.034	0.012	0.034	0.012	1.439	1.103	TRUE	TRUE
	9.910	0.080	0.053	0.010	0.004	0.347	0.229	0.029	0.019	0.023	0.008	0.023	0.008	2.300	1.746	TRUE	TRUE
	9.910	0.080	0.053	0.010	0.004	0.347	0.229	0.029	0.019	0.023	0.008	0.023	0.008	2.300	1.746	TRUE	TRUE
	9.840	0.058	0.040	0.005	0.003	0.254	0.176	0.021	0.015	0.010	0.004	0.010	0.004	1.862	1.457	TRUE	TRUE
	9.840	0.058	0.040	0.005	0.003	0.254	0.176	0.021	0.015	0.010	0.004	0.010	0.004	1.862	1.457	TRUE	TRUE

APPLICATION OF Arc Gis Pro INTO DETERMINATION OF CULVERT LOCATIONS:

Application of Arc Gis Pro into determination of culvert locations

Catchment delineation was carried out to determine the u/s drainage area contributing runoff to the proposed culvert using a 12.5 m resolution Digital Terrain Model (DTM) in ArcGIS Pro. The culvert inlet point (culvert_pts) was used as the pour point for watershed analysis. Hydrological tools including Fill, Flow Direction, and Flow Accumulation were applied to model surface runoff and identify drainage paths. The Watershed tool delineated the catchment area draining toward the culvert, which was then converted to a polygon to calculate its area. Finally, the peak discharge was estimated using the Rational Method .

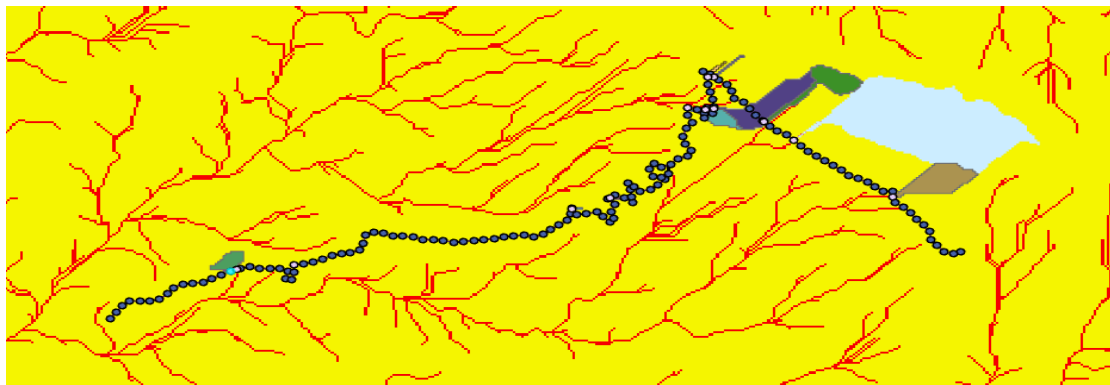


Figure :Catchment Delineation

Impact of Road Widening on existing infrastructures:

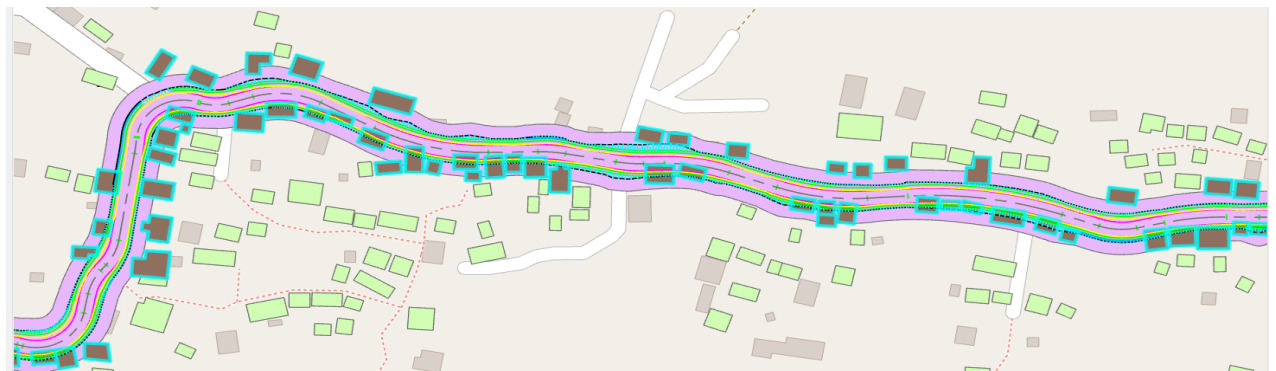


Figure 2 Buildings affected by road widening

An impact analysis was conducted to evaluate the effects of widening the existing single-lane roadway to a double-lane configuration. The analysis primarily focused on identifying existing buildings located within the proposed Right of Way (ROW) that could potentially be affected by the road expansion.

The spatial analysis was performed using ArcGIS Pro by integrating road design data prepared in AutoCAD with geospatial building datasets.

The objective of this analysis was to:

- Identify structures located within the proposed road corridor
- Determine the number of buildings affected by widening
- Provide information for planning compensation, relocation, or design adjustment

The affected buildings are seen as follows:

OBJECT-ID	Name	Type	Shape_Length	Shape_Area
1		school	27.80167497	-45.48773877
2		school	41.14834572	105.5680569
3		school	41.23611988	-89.9746974
4		school	36.21203032	-81.83926856
5		school	24.87272858	-37.47935951
6		school	38.68918582	87.15927702
7	Nepal Red Cross Society, Sudal Sub-Chapter		36.08126392	75.39156411
8	Sudal VDC		59.60586208	212.7516157
9			35.07794616	74.22076475
10			35.50724324	73.08099702
11			32.48665785	63.60217499
12			38.19842023	83.169702
13			39.13932621	89.13875828
14			52.82959492	149.0644426
15		house	34.23405647	-70.19121491
16		house	40.23154786	99.48209031
17		house	33.69815309	70.00531856
18		house	20.85917511	27.11451082
19		house	34.89500074	75.86620053
20		house	22.29768058	30.68937315
21		house	28.98731085	-52.35463106
22		house	26.26402225	-35.81184028
23		house	29.48587093	50.80215933
24		house	25.02036313	38.70315363
25		house	23.87407852	33.92774594
26		house	28.98193581	51.52069378
27		house	32.57132427	58.38442744
28		house	23.67964529	34.47285149
29		house	33.69095398	-66.57711348
30		house	36.99410803	-84.25300741
31		house	27.15151282	-45.90024658
32		house	34.47377374	73.35181372
33		house	28.15384367	47.51515795
34		house	65.60241541	-161.1274387
35		house	35.92537643	76.6123058
36			37.75752189	-87.74559575
37			31.92612081	-62.5677636
38			26.70750885	-44.34599183
39			35.63928521	-79.02849891
40			38.95020676	93.47353295
41			37.860687	-88.21002088
42			37.27491387	-85.30443247
43			41.226693	-95.46551467
44			26.77525999	-44.5008426
45			43.18598148	-108.6970614
46			42.61361323	104.0094565

OBJECT-ID	Name	Type	Shape_Length	Shape_Area
47			34.66440484	-71.84437024
48			37.46175229	-80.04566382
49			36.54919358	83.48357069
50			33.56419755	66.77353431
51			29.9378365	-54.65804408
52			31.77684968	59.9257727
53			39.96400156	94.13892327
54			42.97006292	113.1712321
55			21.13835616	25.25294526
56			55.03194029	158.2148072
57			58.61465792	-159.7785044
58			48.33147366	-135.3341038
59			32.66401955	64.12539523
60			41.38891031	103.169241
61		house	57.72500378	-184.1179185
62		house	49.24366829	-128.3351678
63		house	44.21588181	114.6994554
64		house	34.94270818	-68.19782555
65		house	35.68961507	79.31785436
66		house	35.26660682	69.80398628
67		house	15.8632568	-15.71730695
68		house	42.34104006	-109.1200025
69		house	39.92989802	90.97391209
70		house	30.12025336	-53.62331542
71		house	30.50382535	46.80511715
72		house	38.35690541	-87.71242411
73		house	33.70863271	-63.96324169
74		house	40.51093575	96.59952968
75		house	24.5107484	37.50078384
76		house	34.9428226	-73.85776739
77		house	37.05137595	-83.59229783
78		house	38.40673022	-92.17037517
79		house	45.15989186	109.445748
80			54.08018849	154.5265201
81			43.50794239	116.7532524
82			42.13878891	-87.63798971
83			37.5430214	83.76891287
84			40.76461219	-94.49495751
85			42.88977527	112.4424216
86			35.12213771	70.99004739
87			32.50636108	-62.44403168
88			27.16686772	-46.09715376
89			27.06244884	44.4486899
90			25.12475052	36.32031604
91			27.02127493	-41.82987122
92			25.82662388	-41.55767571
93			38.23063783	-91.21249883
94			41.27466688	-105.5224467
95			49.50517366	151.2422364
96			25.02942921	-39.14961687
97			27.01490531	-44.24567469

OBJECT-ID	Name	Type	Shape_Length	Shape_Area
98			30.48452836	-57.50688289
99			28.00813651	-48.52188006
100			31.83944406	48.25128596
101			34.84644765	75.21503231
102			33.01081383	67.00888212
103			28.29971984	-50.01046832
104			22.61180807	-31.21782482
105			39.34598586	90.26684285
106			38.05084441	-86.98976714
107			37.56918558	85.27679305
108			39.3559024	96.44216737
109			47.56353213	143.2041663
110			33.10731793	66.76871442
111			52.02043467	-167.9143476
112			52.55670256	154.0537743
113			40.47760998	87.85198232
114			32.71273868	66.22690516
115			41.67891896	108.3837137
116			44.30775198	112.7381737
117	Mewa Ko Pasal		54.11447554	161.1167011
118			33.94052947	57.33788229
119			48.51341556	147.0205562
120			43.42077059	-113.5423336
121			35.4681131	-78.13787762
122			46.63512615	-117.3591882
123			51.04153541	128.765011
124			33.41255717	65.80275232
125			29.39428411	53.99724725
126			35.72222558	76.56235262
127			27.70900939	43.85430961
128			37.8768334	-86.86285712
129			45.75185466	-109.0535238
130			32.05141116	61.22575116
131			54.22514377	152.9489908
132			31.06758183	59.00396644
133			41.20417254	-78.7595749
134			32.24745093	62.502886
135			28.19878902	-48.54059881
136			33.34293348	-69.01200601
137			40.04352489	88.099952
138			36.33941556	60.1278424
139			38.04095529	-79.59584665
140			33.37307861	67.53750386
141			40.30720219	95.57465552
142			45.59482998	110.2422521
143			34.1022841	71.49612557
144			36.17211213	-77.81370712
145			33.0669477	67.53712901
146			29.7198804	51.94448656
147			43.02972393	104.2001886
148			37.3368934	86.11922055

OBJECT-ID	Name	Type	Shape_Length	Shape_Area
149			31.49680226	60.1859229
150			26.60465285	42.54844087
151			33.232447	67.29383711
152			69.544573	-264.590842
153			83.20372949	-288.9680045
154			27.31298185	-46.59893339
155			28.97096648	-52.13523241
156			48.37985349	134.6916147
157			59.30435793	-203.1435843
158			37.47806305	-85.28662044
159			45.8211469	130.7570944
160		house	30.86841903	55.8258217
161			33.08931843	67.95497028
162			31.29744436	-60.67755623
163			49.2606845	-143.0125788
164			42.36859961	-106.5248138
165			32.63968443	-60.63987836
166			23.48558978	32.07745551
167			28.07187875	42.93812383
168			28.88109991	51.55013566
169			24.27168396	32.54719322
170			34.49341024	68.85980591
171			31.09416035	59.81466035
172			21.53653451	-28.75042768
173			52.33464126	133.9508246
174			36.02378165	77.25320523
175			48.395168	119.0274464
176			21.08065496	27.76911675
177			24.04709894	29.59948451
178			28.56670033	49.69620913
179			35.08767072	75.95844342
180			35.95735989	80.80746701
181			28.80126243	51.15686173
182	Bindabasini Temple	Place of worship	54.97610805	178.0225756
183			61.23390519	180.8780462
184			25.73342743	-40.64654436
185			35.94012567	-79.62095944
186			29.44215311	52.61801785
187			41.68135775	83.34019311
188			54.88728133	182.9152225
189			42.57771997	93.88536888
190			62.19670954	232.9632503
191			34.49310767	69.20012973
192			25.09556943	37.7656413
193			95.3302211	-303.9774631
194			32.94192198	62.4272818
195			31.94776376	53.87624229